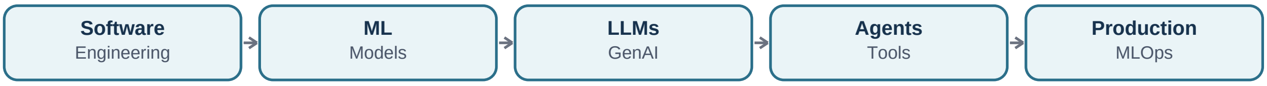
AI Engineering Interview Questions & Answers

80 Questions • Basic to Advanced • Simple English • Practical Examples • Visual Diagrams



Learning path: Foundations → AI system design → ML → MLOps/LLMOps → LLMs → Prompting → Agents → RAG → Architecture → Python backends → APIs → MCP → Cloud → Advanced AI Engineering.

Audio note: interactive PDF audio playback is not included because audio support is inconsistent across common PDF readers. This version prioritizes clear reading and reliable cross-reader compatibility.

Contents

Interview tip: define the concept first, explain why it is used, give one practical example, and mention an important trade-off when relevant.

Part 1 — Core AI Engineering Questions 1–39

The original 39-question foundation is preserved in the previously generated guide. The expanded questions below continue the learning path from Question 40 onward.

Part 2 — Expanded AI Engineering Questions 40–80

40. What is Computer Vision?

Short Answer: Computer Vision is AI that helps computers understand images and video.

Key Points:

• Image classification.

• Object detection.

• Face and image analysis.

Example: Example: Detect cars and pedestrians in a camera image.

41. What is Reinforcement Learning?

Short Answer: Reinforcement Learning learns actions through rewards and penalties.

Key Points:

• Agent.

• Environment.

• Action.

• Reward.

Example: Example: A game-playing agent learns which actions lead to winning.

42. What is a classification problem?

Short Answer: Classification predicts a category or label.

Key Points:

• Binary classification has two classes.

• Multiclass classification has more than two.

Example: Example: Predict whether a transaction is fraud or not fraud.

43. What is regression in Machine Learning?

Short Answer: Regression predicts a continuous numerical value.

Key Points:

• The output is a number.

• The model learns relationships between inputs and the target.

Example: Example: Predict the price of a house.

44. What is precision and recall?

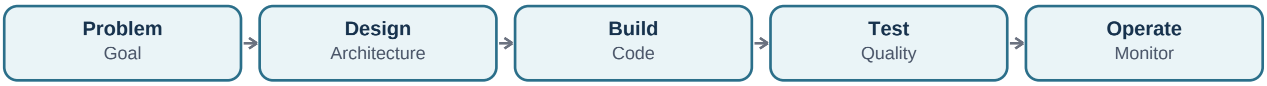
Short Answer: Precision measures how many predicted positives are correct. Recall measures how many real positives were found.

Key Points:

• Precision matters when false positives are expensive.

• Recall matters when missing a positive is expensive.

Example: Example: In fraud detection, missing fraud may make high recall important.



45. What is an ML pipeline?

Short Answer: An ML pipeline is an automated sequence for data preparation, training, evaluation, and deployment.

Key Points:

• Repeatable.

• Versioned.

• Automated.

Example: Example: Data → Clean → Train → Evaluate → Deploy.

46. How do you design an AI system for high availability?

Short Answer: Use redundant services, health checks, retries, timeouts, fallbacks, monitoring, and multiple instances.

Key Points:

• Avoid a single point of failure.

• Use graceful degradation.

Example: Example: If one model provider fails, route requests to a backup provider.

47. How do you handle retries in AI applications?

Short Answer: Retry temporary failures with limits and backoff, but do not blindly retry every error.

Key Points:

• Use exponential backoff.

• Use idempotency for repeated operations.

• Set a maximum retry count.

Example: Example: Retry a temporary network timeout but do not retry an invalid request forever.

48. What is a fallback model strategy?

Short Answer: A fallback strategy uses another model or service when the primary model is unavailable, too slow, or too expensive.

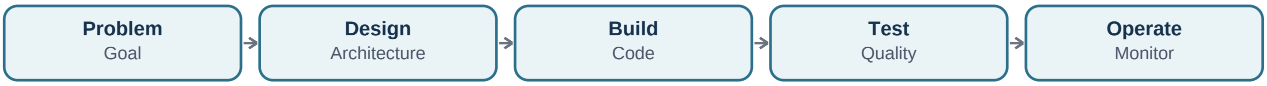
Key Points:

• Improve availability.

• Control cost.

• Maintain user experience.

Example: Example: Primary LLM → backup LLM → safe predefined response.



49. How do you design streaming AI responses?

Short Answer: The backend sends generated output in small pieces as it becomes available instead of waiting for the complete answer.

Key Points:

• Lower perceived latency.

• Better chat experience.

• Requires client and server streaming support.

Example: Example: A user sees an answer appearing word by word.

50. What is model versioning?

Short Answer: Model versioning records which model, data, configuration, and code created a specific result.

Key Points:

• Supports rollback.

• Improves reproducibility.

• Helps compare experiments.

Example: Example: Production uses model v3 while v4 is evaluated safely.

51. What is feature engineering?

Short Answer: Feature engineering transforms raw data into useful inputs for a Machine Learning model.

Key Points:

• Clean data.

• Create meaningful variables.

• Remove irrelevant information.

Example: Example: Convert a timestamp into day-of-week and hour features.

52. What is data validation in ML?

Short Answer: Data validation checks whether incoming data has the expected schema, quality, range, and distribution.

Key Points:

- Detect missing values.
- Detect schema changes.
- Detect unexpected distributions.

Example: Example: Alert when a numeric field suddenly arrives as text.

53. What is an LLM evaluation dataset?

Short Answer: It is a fixed collection of representative questions and expected quality criteria used to compare AI system versions.

Key Points:

- Regression testing.
- Quality comparison.
- Prompt testing.

Example: Example: Every prompt change is tested against 500 known customer questions.

54. What is prompt versioning?

Short Answer: Prompt versioning stores and tracks changes to prompts just like source code.

Key Points:

- Compare results.
- Rollback bad changes.
- Connect prompts to evaluation results.

Example: Example: Prompt v12 improves answer format but increases token usage.

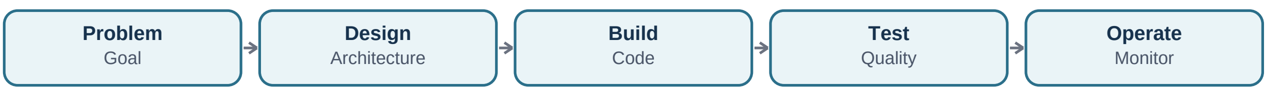
55. What is pretraining?

Short Answer: Pretraining teaches a model general patterns from a very large dataset before it is adapted to specific tasks.

Key Points:

- Large-scale learning.
- General knowledge and language patterns.
- Expensive computation.

Example: Example: A language model learns grammar and broad patterns before instruction tuning.



56. What is instruction tuning?

Short Answer: Instruction tuning trains a model on examples of instructions and desired responses so it follows user requests better.

Key Points:

- Improves instruction following.
- Usually happens after pretraining.

Example: Example: Train examples showing how to answer questions in a specific format.

57. What is temperature in LLM generation?

Short Answer: Temperature controls how random or varied token selection can be.

Key Points:

- Lower temperature is usually more consistent.
- Higher temperature can produce more variety.

Example: Example: Use lower temperature for structured extraction and more variation for creative writing.

58. What is structured output?

Short Answer: Structured output requires the model to return data in a defined format such as JSON.

Key Points:

- Easier validation.
- Better integration with software.
- Reduces parsing problems.

Example: Example: Return {name, email, phone} instead of a paragraph.

59. What is function or tool calling?

Short Answer: Tool calling allows an LLM to request a defined external function with structured arguments.

Key Points:

- The application executes the tool.
- The model should not directly receive unlimited system access.

Example: Example: The model requests get_weather(city='Karachi'), and the backend calls the weather service.

60. What is chain-of-thought reasoning in AI applications?

Short Answer: It refers to internal or structured reasoning steps used to solve complex problems. Production systems should focus on reliable outputs and appropriate evaluation rather than exposing private reasoning.

Key Points:

- Use concise explanations when needed.
- Use tool calls and verification for complex tasks.

Example: Example: An agent can break a task into steps and verify results without exposing hidden internal reasoning.

61. What is prompt chaining?

Short Answer: Prompt chaining splits a complex task into several smaller model calls.

Key Points:

- Each step has a focused purpose.
- Can improve control and debugging.

Example: Example: Extract facts → classify → summarize.

62. What is self-consistency?

Short Answer: Self-consistency compares multiple candidate outputs or reasoning paths and selects a more reliable result.

Key Points:

- Useful for some reasoning tasks.
- Increases cost and latency.

Example: Example: Generate multiple answers and choose the answer that agrees most often.

63. What is an output parser?

Short Answer: An output parser converts and validates model output into a format that application code can safely use.

Key Points:

- Schema validation.
- Error handling.
- Retry or correction.

Example: Example: Convert an LLM response into a validated Pydantic model.

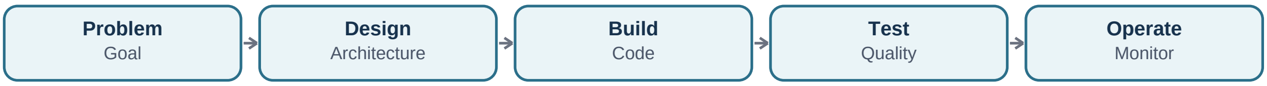
64. What is query rewriting in RAG?

Short Answer: Query rewriting transforms a user's question into a better search query before retrieval.

Key Points:

- Adds missing context.
- Removes unnecessary words.
- Can create multiple search queries.

Example: Example: 'What about it?' becomes 'What is the company's annual leave policy?' using conversation context.



65. What is metadata filtering in Vector Search?

Short Answer: Metadata filtering restricts retrieval using attributes such as user, department, date, or document type.

Key Points:

- Improves relevance.
- Improves access control.

Example: Example: Search only documents belonging to the user's organization.

66. What is RAG evaluation?

Short Answer: RAG evaluation checks both retrieval quality and answer quality.

Key Points:

- Was the correct document retrieved?
- Was the answer supported by the context?
- Was the answer relevant?

Example: Example: A correct answer with irrelevant retrieved documents is still a retrieval problem.

67. What is knowledge graph RAG?

Short Answer: Knowledge Graph RAG combines document retrieval with structured relationships between entities.

Key Points:

- Useful for connected facts.
- Can improve multi-hop questions.

Example: Example: Company → Department → Manager → Project relationships can be queried together.

68. What is a reranker model?

Short Answer: A reranker is a model that scores query-document pairs more precisely after an initial retrieval step.

Key Points:

- Usually more accurate than simple similarity.
- Often used on a smaller candidate set.

Example: Example: Retrieve 100 chunks, rerank them, and keep the top 8.

69. What is agent planning?

Short Answer: Agent planning is deciding the sequence of actions needed to achieve a goal.

Key Points:

- Plan tasks.
- Select tools.
- Check results.
- Adapt when something fails.

Example: Example: Research agent plans search → extract facts → compare sources → summarize.

70. What is human-in-the-loop AI?

Short Answer: Human-in-the-loop means a person reviews, approves, or corrects important AI actions.

Key Points:

- Useful for high-risk decisions.
- Improves safety and trust.

Example: Example: AI prepares a refund but a human approves the final payment.

71. What is agent orchestration?

Short Answer: Agent orchestration controls how agents, tools, data, and workflows communicate and execute.

Key Points:

- Sequential workflows.
- Parallel work.
- Conditional branching.
- Retries and checkpoints.

Example: Example: Research agents work in parallel and a reviewer combines their results.

72. What is an agent loop?

Short Answer: An agent loop repeatedly observes the current state, decides the next action, executes it, and checks the result.

Key Points:

- State.
- Decision.
- Action.
- Observation.

Example: Example: Search → inspect results → search again if information is missing.

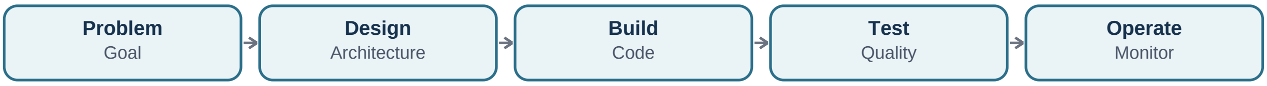
73. What is dependency injection?

Short Answer: Dependency Injection gives a component its dependencies from outside instead of creating them internally.

Key Points:

- Improves testing.
- Reduces coupling.
- Makes replacement easier.

Example: Example: Inject a mock LLM client during tests instead of calling a real provider.



74. What is Pydantic and why is it useful with FastAPI?

Short Answer: Pydantic validates and parses Python data using typed models.

Key Points:

- Request validation.
- Response validation.
- Clear schemas.

Example: Example: FastAPI validates that an AI request contains a valid question and user ID.

75. What is SQLAlchemy?

Short Answer: SQLAlchemy is a Python toolkit and ORM for working with relational databases.

Key Points:

- Database connections.
- Queries.
- Object-relational mapping.

Example: Example: A FastAPI service stores users and conversations in PostgreSQL.

76. What is a virtual environment in Python?

Short Answer: A virtual environment isolates project dependencies so different projects can use different package versions.

Key Points:

- Avoid dependency conflicts.
- Reproducible development.

Example: Example: Project A uses one library version while Project B uses another.

77. What is type hinting in Python?

Short Answer: Type hints describe expected data types and improve readability, tooling, validation, and maintainability.

Key Points:

- Useful with FastAPI and Pydantic.
- Helps catch mistakes earlier.

Example: Example: def add(a: int, b: int) -> int.

78. What is a GPU vs CPU for AI?

Short Answer: CPUs are general-purpose processors. GPUs perform many parallel calculations and are usually better for deep learning inference and training.

Key Points:

- CPU: general backend work.
- GPU: neural network computation.

Example: Example: FastAPI may run on CPU while an LLM inference server uses GPU.

79. What is autoscaling?

Short Answer: Autoscaling automatically increases or decreases resources based on workload.

Key Points:

- Scale API instances.
- Scale workers.
- Scale inference capacity.

Example: Example: Add inference workers during high traffic and reduce them during quiet periods.

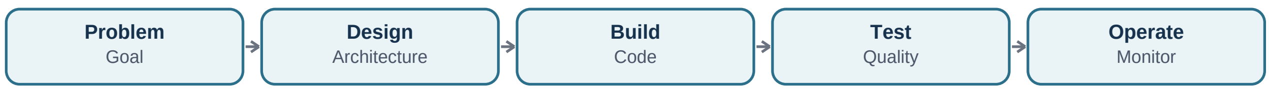
80. What is blue-green deployment?

Short Answer: Blue-green deployment keeps two environments so traffic can switch from the old version to the new version with a quick rollback option.

Key Points:

- Lower deployment risk.
- Easy rollback.

Example: Example: Test the new AI service in green, then switch traffic from blue.



81. What is canary deployment?

Short Answer: Canary deployment sends a small percentage of traffic to a new version before gradually increasing it.

Key Points:

- Detect problems early.
- Reduce blast radius.

Example: Example: 5% of users receive a new prompt/model version first.

82. What is model routing?

Short Answer: Model routing chooses the most suitable model for each request based on complexity, cost, latency, or privacy.

Key Points:

- Simple task → small model.
- Complex task → stronger model.
- Sensitive task → private model.

Example: Example: A classifier routes requests to different models based on task type.

83. What is AI safety in production?

Short Answer: AI safety means reducing harmful, unsafe, unauthorized, misleading, or unreliable behavior.

Key Points:

- Guardrails.
- Permissions.
- Evaluation.
- Monitoring.
- Human review.

Example: Example: An AI assistant cannot execute a sensitive action without authorization.

84. What is an AI gateway?

Short Answer: An AI gateway is a central service that manages access to multiple model providers or model servers.

Key Points:

- Routing.
- Authentication.
- Rate limits.
- Cost tracking.
- Fallbacks.

Example: Example: One internal API routes requests to different LLM providers.

85. How do you debug a bad AI answer?

Short Answer: I inspect the full trace: user input, prompt, retrieved context, tool calls, model version, output validation, and evaluation results.

Key Points:

- First find where the failure happened.
- Then change one part and retest.

Example: Example: If the correct document was not retrieved, improve retrieval before changing the model.

86. How do you choose between an API model and a self-hosted model?

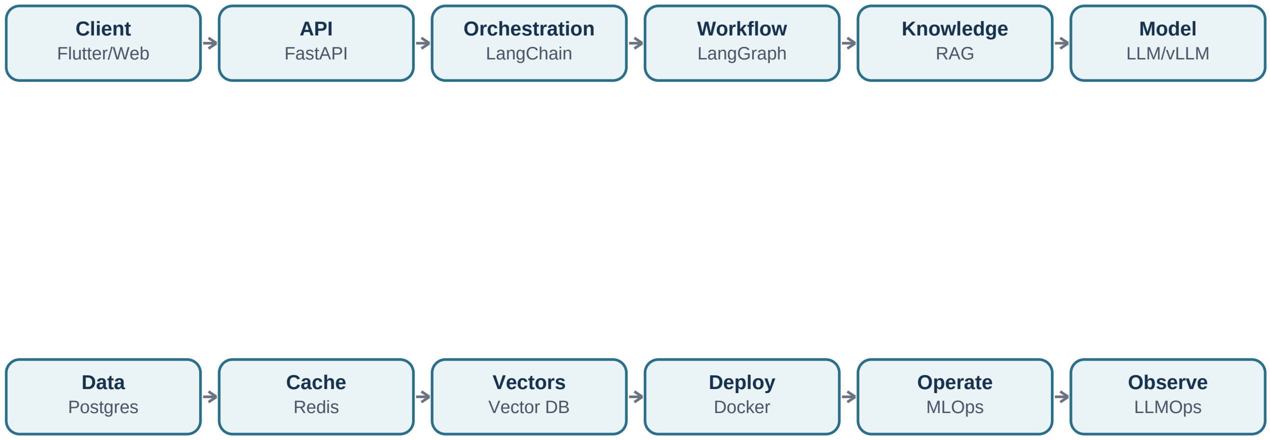
Short Answer: Compare quality, privacy, cost, latency, infrastructure, scaling, and operational complexity.

Key Points:

- API model: faster to start.
- Self-hosted: more control and privacy.
- The best choice depends on requirements.

Example: Example: A regulated private workload may justify self-hosting.

Final AI Engineering Architecture Cheat Sheet



Strong interview answer pattern: Define → Why it is used → Practical example → Trade-offs → Production architecture → Monitoring and improvement.